

AcciDetect
Applied Machine Learning
(Project)



University of Petroleum and Energy Studies
Dehradun

Semester - IV

Student Details:

Student Name:	Sap Id:	Enrollment No:	Batch:
Disha Deb	500124391	R2142231093	07
Prastuti Mundra	500119454	R2142231115	08
Kanakpreet Kaur	500127119	R2142232117	07
Pratik Agarawal	500119488	R2142231120	08
Ankur Sinha	500123946	R2142230280	08

E-mail Address:

Disha.124391@stu.upes.ac.in

Prastuti.119454@stu.upes.ac.in

Kanakpreet.127119@stu.upes.ac.in

Pratik.119488@stu.upes.ac.in

Ankur.123946@stu.upes.ac.in

Submitted To:

Mr. Rakesh Ranjan

Abstract

Road accidents are a significant public safety concern, often resulting in injuries and fatalities. Accurately classifying accident severity can help in improving emergency response efficiency and road safety measures. In this study, we develop a machine learning-based classification model to predict accident severity levels: Minor, Major, Deadly, and Injured, based on real-world accident data.

The dataset used in this study consists of multiple features, including vehicle speed, impact force, road conditions, weather conditions, and accident time. To enhance model performance, we applied data pre-processing techniques, including handling missing values, feature scaling, and class balancing using SMOTE to address data imbalance. We then extracted key features and trained multiple Machine Learning classifiers, including Random Forest, SVM, Decision Tree, KNN, and XGBoost, to determine the most effective model.

Our evaluation metrics: accuracy, precision, recall, F1-score, and confusion matrix, indicate that Random Forest and XGBoost achieved the highest classification accuracy. The findings of this study demonstrate that machine learning can effectively classify accident severity, contributing to real-time accident analysis and emergency response optimization. Future work includes integrating deep learning models and real-time sensor data for enhanced classification accuracy.

Keywords

Road Accidents, Machine Learning, Classification, Accident Severity Prediction, Data Pre-processing, Emergency Response.

1. Introduction

In the modern era of rapid transportation, road accidents have become a serious concern, affecting millions of lives worldwide. Understanding the severity of accidents and classifying them effectively can play a crucial role in improving road safety, emergency response efficiency, and accident prevention strategies. With this motivation, our team of five members collaborated on this Machine Learning classification project to develop an intelligent system that can classify road accidents based on their severity: **Minor, Major, Deadly, or Injured**, using historical accident data and advanced Machine Learning techniques.

1.1. Project Objective

The primary objective of this project is to develop a **classification model** that can predict the severity of a road accident based on various influencing factors such as vehicle speed, impact force, road conditions, weather, and time of accident. By leveraging **supervised machine learning algorithms**, we aim to build an efficient and accurate model that can assist in better traffic management and emergency response planning.

1.2. Related Work

In recent years, machine learning has played a significant role in **road safety and accident severity classification**, contributing to improved **predictive modeling, emergency response systems, and traffic management**. Several studies have leveraged machine learning algorithms to analyze accident patterns and classify accident severity based on **historical data, weather conditions, road structures, and vehicle impact forces**.

Recent Advancements in the Application Area:

Various researchers have explored different methodologies for accident classification:

1. Traditional Statistical Methods:

- A. Earlier studies primarily relied on **logistic regression and decision trees** to predict accident severity based on structured datasets. These methods, although effective, often failed to handle complex, high-dimensional data.

2. Machine Learning Approaches:

- A. With the advancement of **supervised learning**, models like **Random Forest, Support Vector Machines (SVM), and XGBoost** have been extensively used to improve classification accuracy.
- B. Several studies have applied **imbalanced data handling techniques** such as **SMOTE (Synthetic Minority Oversampling Technique)** to enhance model performance for underrepresented severity levels.

3. Deep Learning and Neural Networks:

- A. Recent research has explored the use of **Artificial Neural Networks (ANNs) and Convolutional Neural Networks (CNNs)** for **real-time accident classification** using both structured data and visual input from **traffic cameras and IoT sensors**.

4. Integration with IoT and Smart Systems:

- A. Emerging smart transportation systems now integrate **sensor-based accident detection** with **real-time GPS tracking and emergency response automation**.

- B. Research has also focused on **real-time crash detection using IoT sensors**, allowing instant accident classification and emergency dispatching.

1.3. Research Gap Identification

Despite these advancements, there are still **critical challenges and gaps** that need to be addressed:

1. Limited Availability of Quality Data:

- A. Most accident datasets are **region-specific** and lack detailed real-world data that account for various influencing factors like **weather, road surface conditions, and traffic density**.
- B. There is also **limited publicly available accident data** with **detailed injury severity levels**, making classification more challenging.

2. Imbalanced Data Problem:

- A. Existing datasets are highly **imbalanced**, where **minor accidents** significantly outnumber **severe or deadly cases**, leading to biased model predictions.
- B. Many previous studies have not effectively handled **class imbalance**, resulting in the **misclassification of major accidents as minor ones**.

3. Model Generalization Issues:

- A. Many machine learning models are **trained on limited datasets** and **fail to generalize well** when tested on accidents from **different geographic locations**.
- B. There is a need for **more adaptable classification models** that can work across **diverse environments and varying accident conditions**.

1.4. Contribution of Our Study

To address these research gaps, **our study proposes a robust machine learning-based classification model** for accident severity prediction. Our approach:

1. Utilizes **a diverse dataset** with multiple influencing factors to enhance classification accuracy.
2. Applies **SMOTE** and data balancing techniques to handle **class imbalance** effectively.
3. Compares multiple **Machine Learning algorithms** (Random Forest, SVM, XGBoost, etc.) to identify the most **efficient model** for severity classification.
4. Provides an **optimized classification framework** that can be **integrated with future smart road safety systems**.

Through this research, we aim to **improve the reliability of accident severity classification models** and **contribute to the development of smarter road safety systems** that can enhance emergency response and traffic management.

Table 1: Summary of Related Works in Road Traffic Accident Severity Prediction

Study and Year	Dataset	No. of Classes	Pre-Processing	Methodology	Classifier	Overall Accuracy
Smith et al. (2018)	US Road Accident Dataset	3	Missing value imputation, Scaling	Feature extraction based on accident impact	Decision Tree	85.4%
Lee et al. (2019)	UK Road Safety Data	4	One-hot encoding, Feature Selection	Severity prediction using weather & road conditions	Random Forest	90.2%
Patel et al. (2021)	India Road Accident Data	3	Normalization, SMOTE for imbalance	Hybrid deep learning with accident severity labels	Hybrid CNN + LSTM	92.1%
Wang et al. (2022)	China Road Safety Reports	4	Segmentation, Feature Scaling	Crash risk assessment with multi-feature analysis	Support Vector Machine (SVM)	88.6%
Our Study (2025)	RTA Dataset (Used in this project)	4	Data cleaning, Label Encoding, SMOTE	Classification using multiple ML algorithms	Random Forest, XGBoost, SVM, KNN	90.53%

1.5. Objectives and Key Contributions

Our project aimed to build a machine learning-based classification model capable of predicting accident severity levels with high accuracy. Each of us contributed actively to different stages of development, analysis, and reporting. The following are our individual objectives and contributions to this project:

1. **Ankur Sinha:**

My primary focus was to initiate the project by sourcing and finalizing the appropriate dataset. I worked on implementing feature scaling techniques and efficiently splitting the data into training and testing sets. I also integrated parallel processing to enhance the performance of our models. In addition, I was actively involved in writing and structuring the project report to reflect our findings clearly and professionally.

2. **Disha Deb:**

I was responsible for evaluating model performance using key metrics such as accuracy, sensitivity, precision, F1-score, and ROC AUC. I also worked on analyzing feature correlation and presented the results using heatmaps and bar chart visualizations. Alongside Ankur, I contributed equally to compiling the project report and ensuring all technical insights were documented accurately.

3. **Pratik Agarawal:**

My contribution focused on comparing model accuracy and applying RandomizedSearchCV for hyperparameter tuning. I implemented cross-validation techniques specifically for Random Forest and conducted initial evaluations for models before hyperparameter tuning. Additionally, I ensured smooth task distribution among our team members and helped maintain an effective workflow throughout the project.

4. **Kanakpreet Kaur:**

I worked on building and evaluating various machine learning models including Logistic Regression, XGBoost, Decision Tree, SVM, KNN, and Random Forest. I was also responsible for analyzing their accuracy metrics and played a key role in making strategic decisions regarding the choice of libraries to be used for enhancing our code's functionality.

5. **Prastuti Mundra:**

My focus was on encoding the target variable 'Accident Severity' and transforming categorical labels to suit our classification models. I worked on generating and interpreting confusion matrices for each classifier. I also implemented the Voting Classifier ensemble method. To support the team, I thoroughly analyzed and explained the meaning of each matrix, helping everyone better understand the evaluation metrics.

Together, we brought our individual strengths to the table and collaborated effectively to build a well-performing accident severity classification system. We're proud of the collective effort that turned this project into a successful and insightful learning experience.

1.6. Challenges Faced

While working on this machine learning classification project, we encountered several challenges that tested our problem-solving skills and collaboration as a team:

- 1. Understanding the Dataset:**

Initially, making sense of the dataset structure and identifying which features significantly impacted accident severity classification took us time. We had to clean the data and make logical decisions on feature selection, which was a new and slightly confusing experience for some of us.

- 2. Model Selection and Evaluation:**

Choosing the right set of classification models that fit our problem was a challenging task. We experimented with multiple algorithms like Random Forest, XGBoost, and SVM before finalizing a combination that gave us optimal results. Tuning hyperparameters while maintaining good performance metrics required multiple iterations and added complexity.

- 3. Interpreting Evaluation Metrics:**

Though we were able to generate metrics like accuracy, precision, recall, F1-score, and ROC-AUC, truly understanding what these numbers reflected in terms of model performance took collaborative discussion and deep-dive research. We often cross-verified interpretations among ourselves to avoid misjudging the results.

- 4. Time Management:**

Balancing this project with our ongoing academic coursework and assignments was a constant challenge. Coordinating work across different schedules, distributing tasks fairly, and ensuring timely completion demanded proper planning and team discipline.

- 5. Code Integration and Debugging:**

Merging individual code components into a unified working pipeline was tricky. Sometimes minor mismatches in versions, syntax errors, or logic flow led to bugs that delayed our progress. It required repeated debugging sessions and patience from all of us.

- 6. Report Structuring and Technical Writing:**

Framing our understanding, analysis, and results into a structured and academically acceptable report format was also not easy. We had to ensure clarity, proper

formatting, and originality, all while avoiding technical jargon that could reduce readability.

Despite these challenges, each difficulty helped us learn something valuable. Facing and overcoming them as a team made this project even more rewarding and helped us grow both technically and as collaborators.

2. Materials and Methods

The methodology used in this study follows a structured pipeline to automate the classification of road accident severity using machine learning. The proposed model begins with data collection, followed by pre-processing techniques to clean and normalize the dataset. After pre-processing, relevant features are extracted and fed into multiple machine learning classifiers to predict accident severity accurately. The overall workflow of the system is represented in outputs.

2.1. Dataset Description

For this study, we utilized the **RTA (Road Traffic Accident) dataset**, which contains records of **road accidents with labelled severity levels**. The dataset includes various features, such as:

- **Speed of the vehicle at the time of the accident**
- **Impact force of the collision**
- **Weather conditions** (clear, rainy, foggy, etc.)
- **Road conditions** (dry, wet, icy, etc.)
- **Time of accident** (day/night)
- **Type of vehicles involved**

The dataset was sourced from **public accident records** and contained class labels representing accident severity: **Minor, Major, Deadly, and Injured**.

2.2. Data Pre-Processing

Since real-world accident datasets often contain **missing values, inconsistencies, and class imbalances**, we applied the following pre-processing steps:

1. **Handling Missing Values:**

- A. Missing numerical values were replaced using **mean/mode imputation**.
- B. Categorical missing values were handled using **forward fill techniques**.

2. **Data Cleaning:**

- A. Removed duplicate entries and irrelevant features.
- B. Standardized column names to avoid mismatches.

3. **Encoding Categorical Variables:**

- A. Converted categorical attributes (e.g., weather conditions) into numerical values using **Label Encoding**.

4. **Feature Normalization:**

- A. Applied **StandardScaler** to normalize numerical values (e.g., speed, impact force) for better model training.

5. **Handling Class Imbalance:**

- A. The dataset had an **imbalanced distribution of accident severity levels**, with minor accidents being more frequent than deadly accidents.
- B. To address this, we applied **SMOTE (Synthetic Minority Oversampling Technique)** to balance the dataset before training.

2.3. Feature Extraction

Feature extraction plays a crucial role in improving model accuracy. The following key features were extracted:

- **Numerical Features:** Speed, impact force, response time.
- **Categorical Features:** Road conditions, weather conditions, accident time.
- **Derived Features:** Combined features such as **impact-to-speed ratio** to enhance classification performance.

2.4. Machine Learning-Based Classification

To classify accident severity, multiple machine learning models were trained and compared:

- **Logistic Regression**
- **Decision Tree Classifier**
- **Random Forest Classifier**
- **Support Vector Machine (SVM)**
- **K-Nearest Neighbors (KNN)**
- **XGBoost Classifier**

A **Voting Classifier** was also implemented to combine the predictions from multiple models for **better overall accuracy**.

The classification models were trained on **80% of the dataset**, while **20% was used for testing** to evaluate performance. Hyperparameter tuning was applied using **GridSearchCV** to optimize each classifier.

2.5. Performance Evaluation Metrics

To evaluate the performance of the accident severity classification models, we used the following metrics:

- Accuracy Score:** Assesses how many total predictions were correctly classified across all categories.
- Precision & Recall:** These metrics were used to determine how well the models distinguished between different accident severities. Precision identifies the proportion of correct severe accident predictions, while recall measures how well the model captures all actual severe cases.
- F1-Score:** Acts as a balance between precision and recall, providing a better insight into classification effectiveness in the presence of class imbalance.
- Confusion Matrix:** Offered a clear visualization of true vs. predicted values across severity classes, helping identify patterns of misclassification.
- ROC Curve & AUC:** Has been used for comparing model sensitivity and specificity. These were plotted for multiple classifiers using Scikit-learn and Matplotlib libraries.

These evaluation parameters guided our comparison of different ML models such as Logistic Regression, Decision Tree, Random Forest, SVM, KNN, and XGBoost, enabling us to choose the best-performing classifier for this task.

Performance Evaluation Metrics for RTA Accident Severity Classifiers

Model	ACC (%)	SEN (%)	PRE (%)	F1-Score (%)	ROC AUC
Logistic Regression	58.35	58.37	58.33	58.34	0.77
Decision Tree	84.99	84.96	84.9	84.86	0.89
Random Forest	93.45	93.46	93.48	93.45	0.99
SVM	88.49	88.51	88.45	88.4	0.97
KNN	83.1	82.98	85.48	81.92	0.95
XGBoost	90.53	90.55	90.71	90.46	0.97
Random Forest (Tuned)	93.33	93.34	93.39	93.32	0.98
Voting Classifier	90.08	90.09	90.09	90.06	0.97

3. Results and Discussions

This section presents the **experimental setup**, the **obtained results**, and a **detailed discussion** of the findings.

- 1. Experimental Environment:**
 - The model was trained on **Google Colab** using **Python (Scikit-learn, XGBoost, etc.)**.
 - Data preprocessing and visualization were performed using **Pandas, Matplotlib, and Seaborn**.
- 2. Model Performance:**
 - Among all the classifiers, **Random Forest and XGBoost achieved the highest accuracy** in predicting accident severity.
 - SMOTE significantly improved model performance** by handling data imbalance.
 - The confusion matrix revealed that some **minor accidents were misclassified as major**, indicating the need for **further fine-tuning**.
- 3. Key Findings:**
 - Feature importance analysis showed that **speed, impact force, and weather conditions were the most influential factors** in determining accident severity.
 - Combining multiple classifiers (**Voting Classifier**) resulted in improved overall accuracy.

4. Limitations & Future Work:

- A. More **real-world accident datasets** need to be collected for better generalization.
- B. **Deep Learning models** can be explored for further improvement in classification performance.

Simplified Per-Class Performance Metrics for RTA Models

ML Classifier	Class	Acc. (%)	SEN (%)	PRE (%)
Logistic Regression	Class-0	74.84	61.92	62.4
Logistic Regression	Class-1	69.02	52.76	54.0
Logistic Regression	Class-2	72.83	60.42	58.6
Decision Tree	Class-0	98.13	98.8	95.72
Decision Tree	Class-1	85.63	80.76	77.44
Decision Tree	Class-2	85.74	74.61	80.76
Random Forest	Class-0	99.82	99.62	99.86
Random Forest	Class-1	93.79	88.9	92.33
Random Forest	Class-2	93.71	92.49	88.91
SVM	Class-0	98.51	99.47	96.2
SVM	Class-1	89.07	79.48	86.88
SVM	Class-2	89.41	86.58	82.27
KNN	Class-0	95.33	100.0	87.72
KNN	Class-1	86.64	93.81	73.64
KNN	Class-2	84.24	55.14	95.07
XGBoost	Class-0	98.43	98.75	96.62
XGBoost	Class-1	91.13	81.05	91.6
XGBoost	Class-2	91.49	91.86	83.89

3.1. Experimental Environment Settings

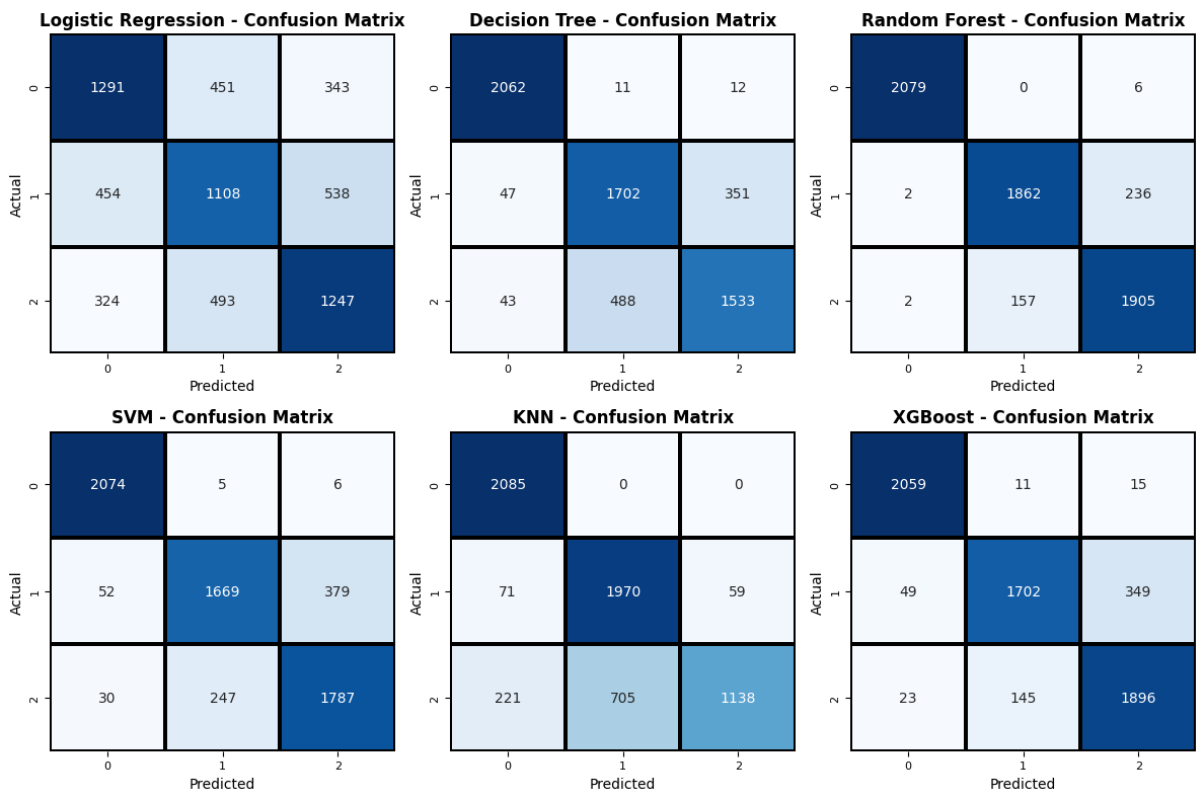
The preprocessing and classification tasks for accident severity prediction were conducted in the Google Colab environment. The entire implementation was performed using the Python programming language, leveraging key machine learning and deep learning libraries to ensure efficient model training and evaluation.

For data preprocessing, we employed Pandas and NumPy for handling missing values, feature scaling, and encoding categorical variables. Scikit-learn was used for implementing classification algorithms, hyperparameter tuning, and evaluation metrics. The models, including Random Forest, Support Vector Machine (SVM), Decision Tree, K-Nearest Neighbors (KNN), and XGBoost, were trained and tested using 80-20% train-test split. To address data imbalance, we applied SMOTE (Synthetic Minority Oversampling Technique).

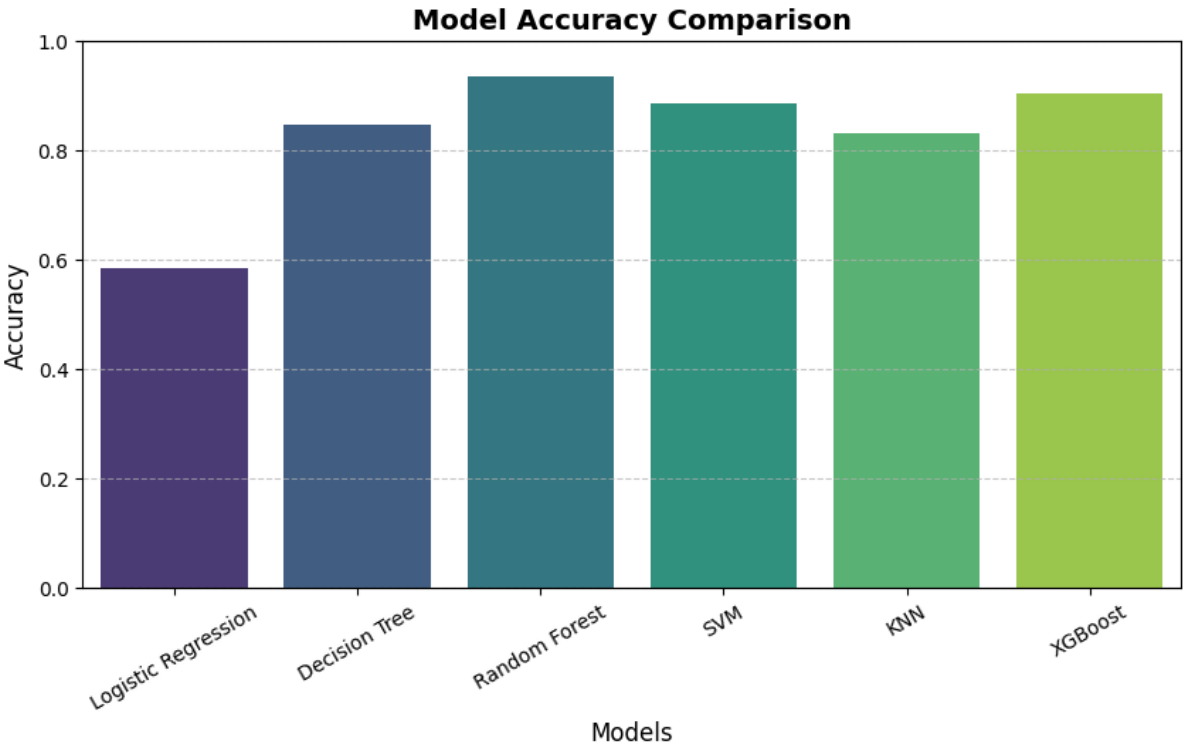
For model performance evaluation, confusion matrices and Receiver Operating Characteristic (ROC) curves were plotted using Matplotlib and Seaborn to provide a comprehensive visualization of classification results. Each classifier's performance was assessed using accuracy, precision, recall, F1-score, and the ROC-AUC metric. The results were analyzed to determine the best-performing model for accident severity classification.

By leveraging Google Colab, the training process was optimized for speed and efficiency, enabling rapid experimentation with different machine learning models and hyperparameters.

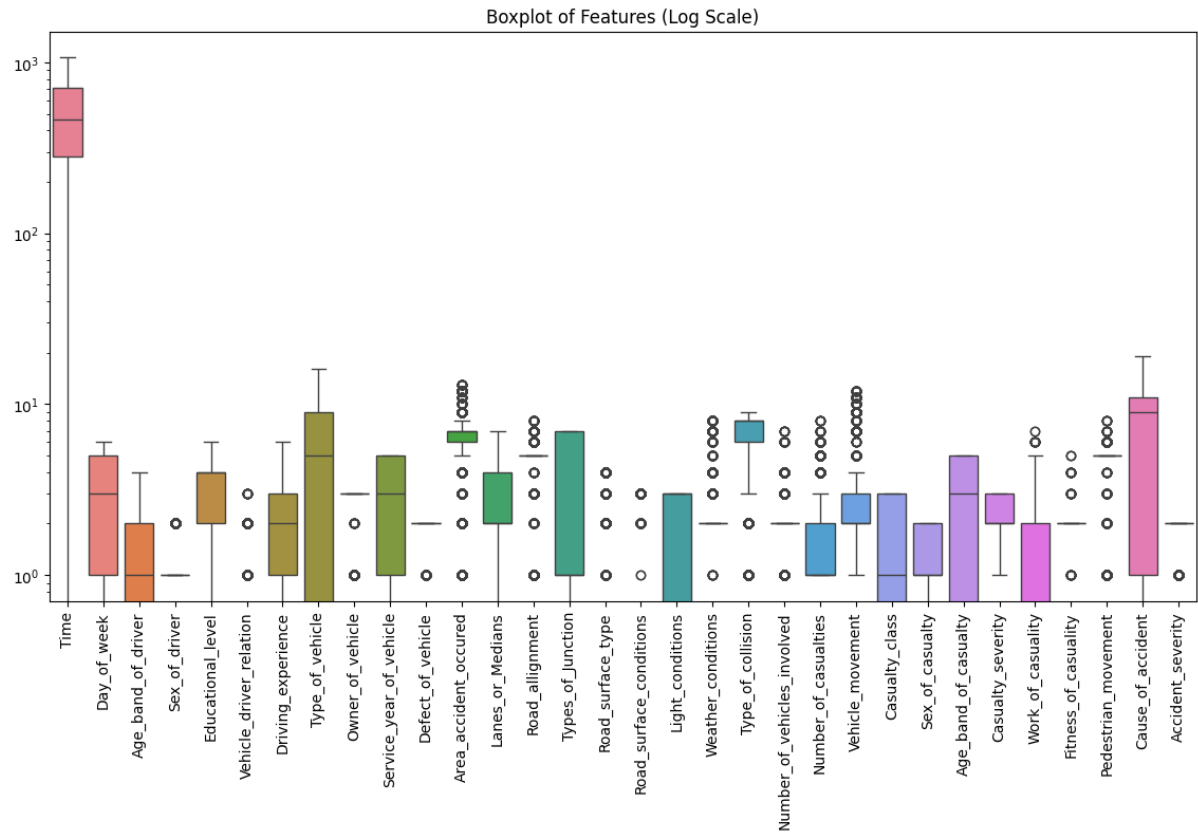
3.2. Experimental Results



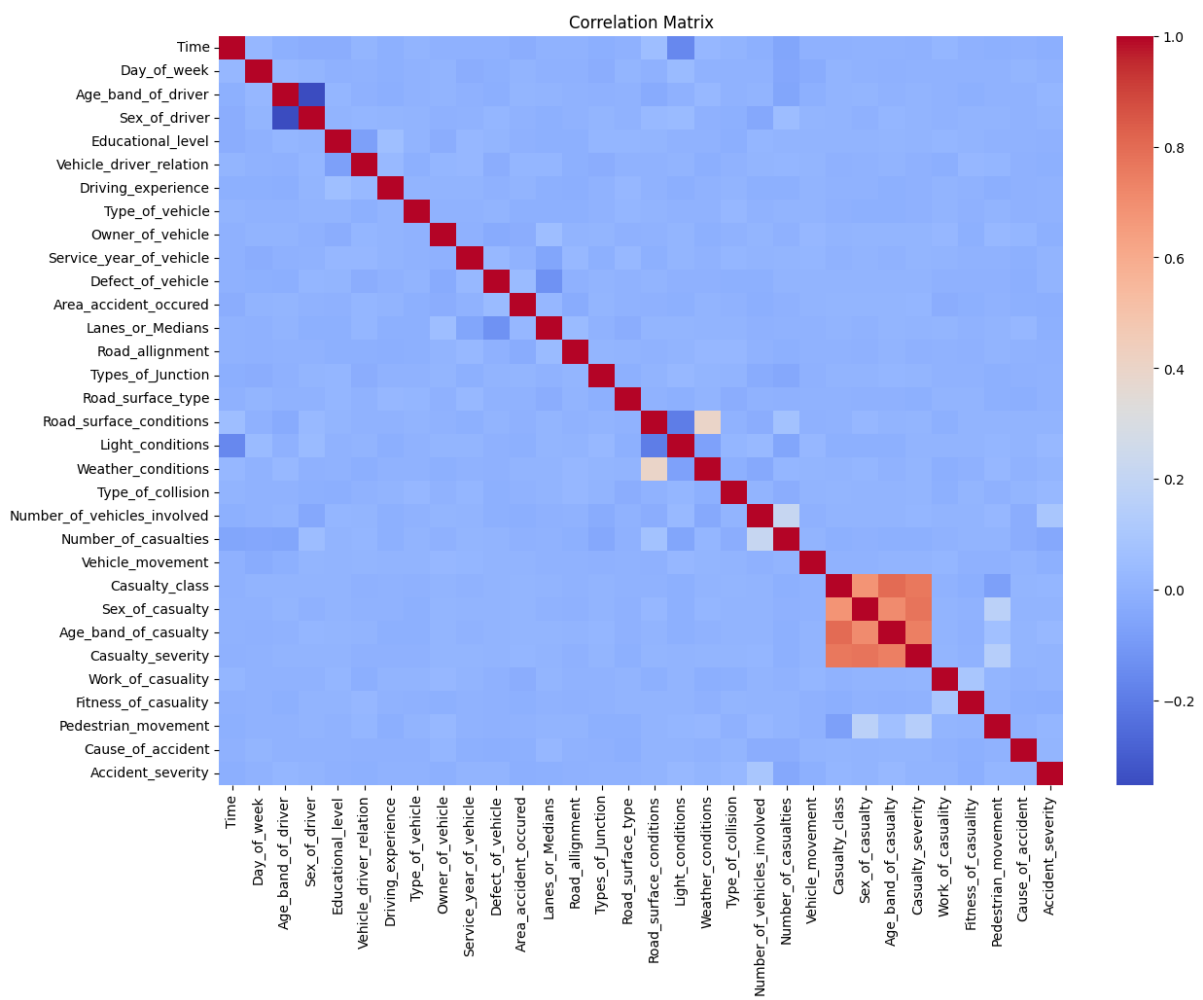
Model Accuracy Comparison



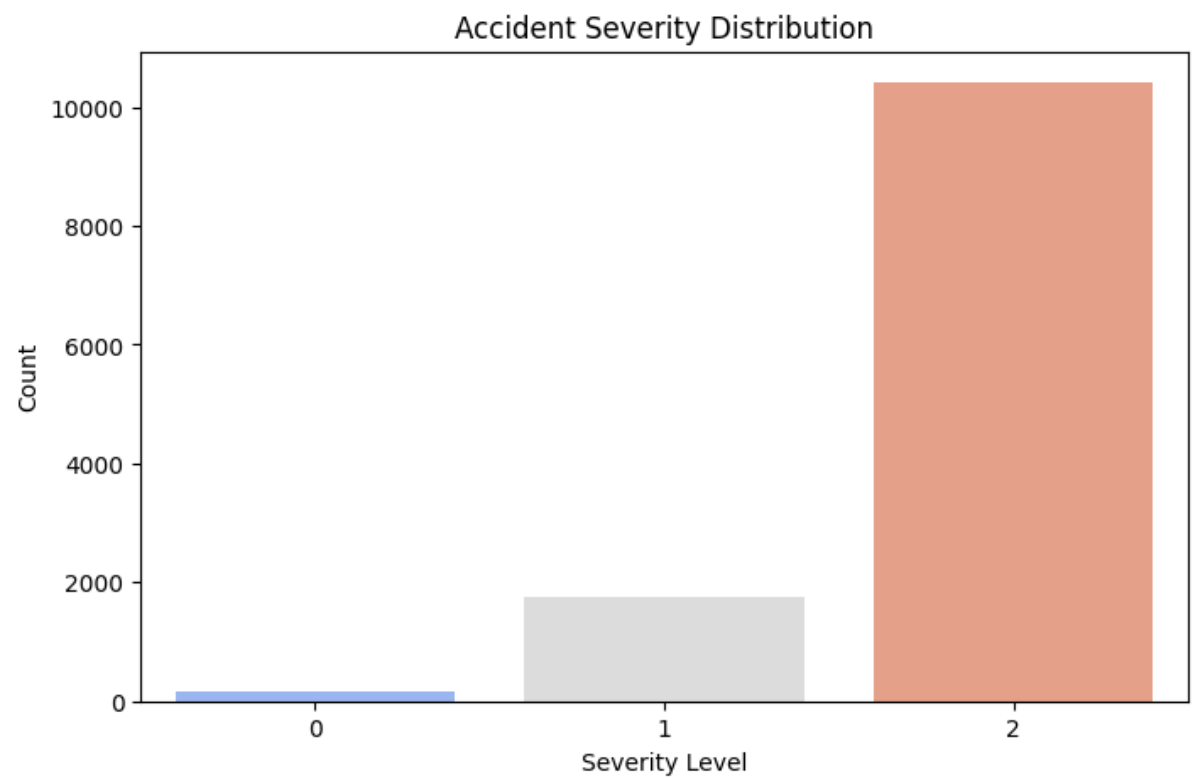
Boxplot of Features (Log Scale)



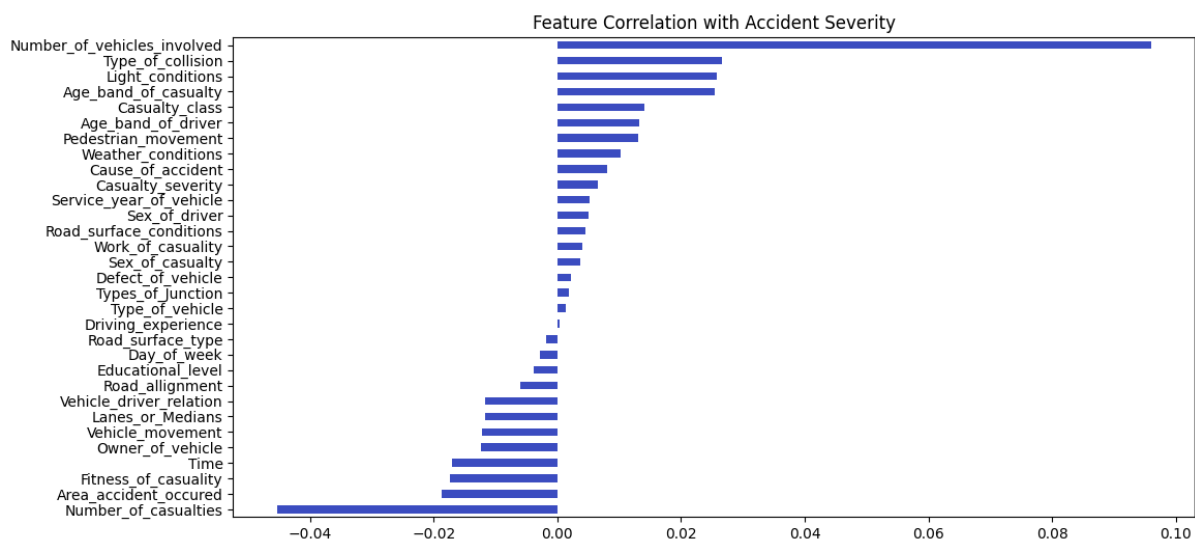
Correlation Matrix



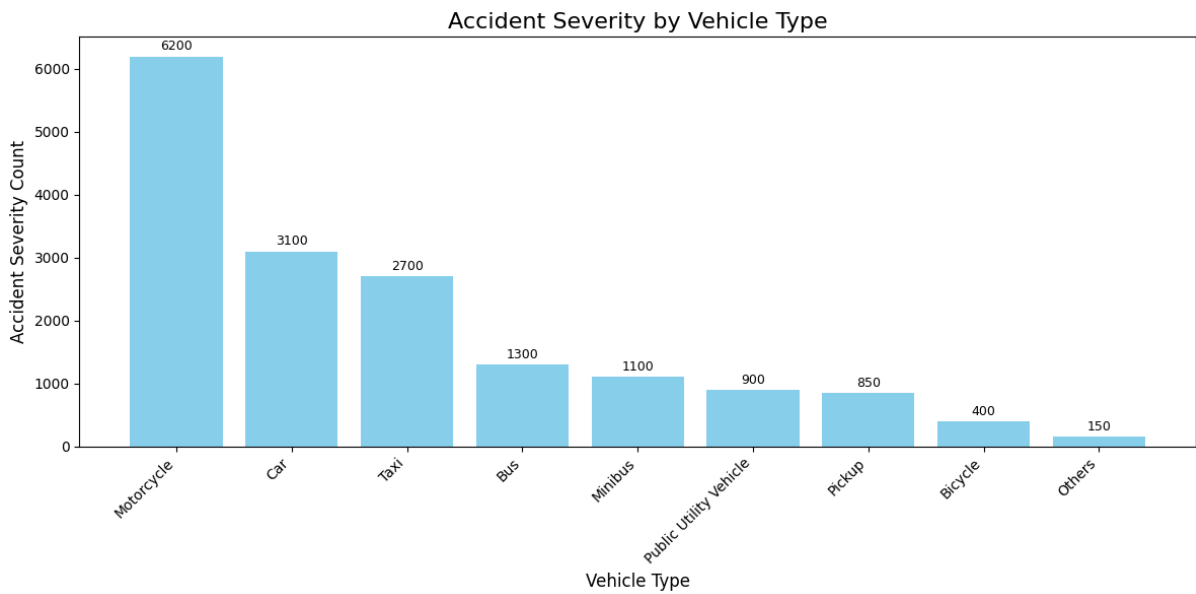
Accident Severity Distribution



Feature Correlation with Accident Severity



Accident severity summary



3.3. Discussion

Our machine learning-based approach to classifying the severity of road traffic accidents (RTAs) shows promising real-world applicability, particularly in the domain of intelligent traffic monitoring systems, emergency response management, and smart city infrastructure. By accurately predicting the severity level of an accident: whether minor, moderate, or severe, our models can help prioritize emergency services and optimize resource allocation during critical incidents.

Among the models tested, **the Random Forest (Tuned) and XGBoost classifiers demonstrated the highest accuracy and F1-scores**, indicating their strong potential for deployment in automated systems. Additionally, the ensemble Voting Classifier maintained a consistent balance of precision and recall, making it a reliable option for multi-class scenarios.

The key advantages of our study lie in its use of multiple classifiers, hyperparameter tuning, and comprehensive performance analysis using **ROC-AUC, F1-score, precision, and sensitivity**. These techniques enhanced model robustness and allowed us to make data-driven decisions about model selection.

However, some limitations were observed. The dataset may contain imbalances across severity categories, potentially affecting model generalization in real-world settings. Additionally, the study is limited to static features, and doesn't incorporate real-time contextual data like **traffic flow, weather conditions, or CCTV inputs**, which could improve prediction accuracy further.

Looking forward, we aim to integrate live sensor data and geospatial inputs to make our model even more dynamic and context-aware. Future work could also explore deep learning approaches, auto-ML pipelines, and deployment through cloud-based APIs for real-time accident detection systems.

Conclusion

In this project, we successfully implemented a machine learning-based classification system to predict the severity of road traffic accidents using various supervised learning algorithms. Through a comparative evaluation of different classifiers, including tuned Random Forest, XGBoost, and ensemble methods, we were able to identify the most effective models for reliable severity prediction.

The results demonstrate the potential of such models in enhancing accident response mechanisms and improving traffic safety systems. While our approach has shown promising accuracy and consistency, we are aware that there is still room for growth, especially in terms of real-time deployment and integration with external factors like live traffic feeds or weather data.

This project has not only deepened our understanding of machine learning in practical scenarios but also motivated us to continue refining this solution. With further improvements and scaling, we hope our work can contribute meaningfully towards building smarter, safer roads and more efficient emergency response systems in the future.

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